Don Fox
donfox1@mac.com | (865) 360-0089 | 896 Hereford Dr. | Akron, OH 44303

Summary:

Software engineer with a background in telecommunications, warehousing, healthcare IT, and data systems. Most of my work has been in background development using C/C++, Python, Perl, and Elixir, with a focus on RESTful APIs, database design in Unix/Linux environments.

I've mentored interns and worked on collaborative projects, over many occasions. Currently exploring AI engineering and functional programming with Elixir and Haskell. Available for remote or freelance opportunities.

Education:

- **Bachelor of Arts in Philosophy, 1982**
  University of Tennessee, Knoxville TN

Technical Training

- **Associate of Science Degree in Computer Science Technology, *Cum Laude,* 1991**
  Pellissippi State Technical Community College, Knoxville TN

Technical Skills

Languages: Python, C, C++, Perl, Elixir, Haskell, PL/SQL, C#
Frameworks: Django, Flask, Phoenix, SQLAlchemy,
Databases: PostgreSQL, MySQL, Oracle, MongoDB, MariaDB, Redis, Informix, DB2
Platforms: Linux, Unix, macOS, Solaris, AIX, Windows
Tools & Technologies: Git, Docker, Node.js, Shell Scripting, RESTful APIs.

Professional History:

EMRTS
https://emrts.us/

Freelance developer for a healthcare IT startup across six engagements, focused on open-source solutions for medical records, billing, provider enrollment and claims. Projects included rapid prototyping and R&D with academic and professional collaborations.

Note: Gaps between EMRTS engagements are covered by contracts listed below.

- **July 2025-Present:** Developing a proof-of-concept application for a Provider Vault medical enrollment website. Combines Elixir/Phoenix for scalability and concurrency with Python components for AI prompting to explore how large language models can augment provider data workflows. Mentoring interns on complementary healthcare IT projects and supplementing rapid prototyping. Tech: Elixir, Phoenix, Python 3.x,
- **August 23, 2024 – December 2024:** Collaborated with interns on a Medicaid prediction research project, including foundational data science concepts and workflows. Led a follow-up imitative exploring cryptocurrency. Developing a Bitcoin indexer prototype. Project paused when students returned to academic commitments. Tech: macOS, Windows, EMRTS Linux network, GitHub, Python 3.x, Flask, Postgres.
  - *(Interim: Cspire contract July 2018-jam2019; Orchid Orthopedic Solutions, Feb 2020-Dec 2020; Quantuvos, Nov 2021 Feb-2022)*
- **July 2015 – January 2017:** Developed proof-of-concept tools and prototypes for processing medical transaction data (EDI_X12_837) and handling large medical provider datasets (JSON, XML). Extended a web services application for state governments and clinics to facilitate medical provider data registration and validation. **Tools:** Python 3.5, PostgreSQL, Django, JavaScript | **Platform:** Unix
- **October 2013 – March 2015:** Developed research-driven projects for mobile medical screenings and provider enrollment for state governments. Contributed to developing a Health Information Exchange(HIE) application for hospitals. **Tools:** Elixir/Phoenix | **Platform:** Ubuntu Linux, macOS | **Databases:** MariaDB, PostgreSQL, MongoDB
  - (*Interim: Hilton contract Apr 2012 Aug 2012 and Lowes Inc., Aug 2012-Jun2013)*
- **December 2011 – March 2012:** Focused on electronic medical records, claims processing, and provider enrollment applications on GNU/Linux. Developed tools for handling JSON data and extended the application's support for additional records processing.
  Tools: Python, Django, JavaScript (jQuery, Node.js) | **Databases:** MariaDB, PostgreSQL, Redis
  - *(Interim: Hewlett-Packard contract May 2010-Dec 2011)*
- **July 2009 – May 2010:** Conducted code analysis for GNUmed, an open-source electronic medical record system, to evaluate its suitability for client needs. Built a RESTful web service for state medical provider validation. **Tools:** Python, Django | **Platform:** GNU/Linux

Quantuvos

Remote | Nov 2021 – Feb 2022

Technical consulting for Quantuvos, a coaching-platform startup. Reviewed and tested code for the MVP used by coaches and clients to schedule workshops and sessions. Performed code reviews, fixed issues, and recommended improvements to functionality. **Tech:** Python 3.9, Django 3.1, PostgreSQL 14.1, Docker

Orchid Orthopedic Solutions

Remote | Feb 2020 – Dec 2020 https://www.orchid-ortho.com

Contract for document processing and archival, preparing Word, Excel, and PDF files for submission to EtQ Reliance. Modified Word documents by inserting tags and formatting for archival requirements. Programmatically manipulated spreadsheet metadata to verify accuracy and completeness. **Tech:** Python 3.8, C# | Windows/macOS

CSpire

Jackson, MS | July 2018 – Jan 2019 https://www.cspire.com

Developed a RESTful auto-provisioning API for managing customer and vendor product and services data, tracking, logging, and reporting on customer services provided by CSpire and third-party vendors. The application supported both the business office and network system administrators. Contract position. **Tech:** Python 2.7, Flask, SQLAlchemy, MySQL, PostgreSQL

Lowes Inc.

Mooresville, NC | Aug 2012 – Jun 2013 https://corporate.lowes.com

Refactored and extended legacy C code to support the transition to EMV credit and debit card transactions. Tasks included code analysis, bug fixes, and documentation. Contract position.**Tech:** C, AIX, DB2, Oracle

Hilton

Memphis, TN | Apr 2012 – Aug 2012

Provided production support for Hilton's Central Reservation System. Contributed programming support using C, Perl, and shell scripting. **Tech:** Solaris, GNU/Linux, Informix, MySQL

Hewlett-Packard

Nashville, TN | May 2010 – Dec 2011

Performed software maintenance and troubleshooting for TennCare (State of Tennessee). Key projects included:

- Refactored and extended a SOAP server (written in C) for enhanced data exchange within the Medicaid Management Information System (MMIS).
- Developed monthly report generators for medical claims processing.
- Ported the entire application codebase to a new 64-bit Solaris network with updated compilers and Oracle database version.
- Contributed to technical writing and documentation.

Tech: C, PL/SQL, SOAP, Oracle, Solaris

ACS

Nashville, TN | Feb 2009 – Jun 2009

Migrated MMIS (Medicaid Management Information System) software for TennCare from Hewlett-Packard's infrastructure to ACS. Developed automation scripts and analyzed legacy C modules. **Tech:** C, Python 3, Unix

Nationwide

Columbus, OH | Oct 2008 – Feb 2009 Analyzed annuity and financial data daily to detect anomalies in high-volume datasets. Proposed and developed a Python-based data validation and reporting tool to modernize and replace aging shell scripts used in production. **Tech:** Solaris, Oracle, proprietary scripting

Lexmark

Lexington, KY | May 2007 – Oct 2008 https://www.lexmark.com/en_us.html

Performed bug fixes, code analysis, and C code refactoring for high-end prototype printers designed for secure environments (government, military, finance). Developed Python scripts for daily integration testing. Contract position. **Tech:** C, Python 2.6, RHEL

IBM Internet Security Systems

Atlanta, GA | Jan 2007 – May 2007

Developed tools for porting and rebuilding C/C++ applications originally developed in Microsoft Visual Studio for deployment in Unix environments. The approach leveraged residual artifacts generated by Visual Studio, rather than relying on traditional makefile-based builds. Contract position. T**ech:** C/C++, macOS, Ubuntu Linux, Windows XP

Lowe's Inc.

Wilkesboro, NC | Feb 2006 – Dec 2006

Internationalized software for Lowe's first Canadian store. Maintained and extended existing C and shell scripting on AIX.

Tech: C, AIX, Oracle, DB2

Cingular Wireless

Alpharetta, GA | Aug 2004 – Feb 2006

Maintained and extended a legacy billing engine running on AIX and GNU/Linux using Oracle. The system was written primarily in C++ with shell scripts. Taught a C++ course to the code acceptance testing group and developed a Perl application that automated the configuration of their test boxes. **Tech:** C, C++, Perl, Oracle, AIX

SAIC

Oak Ridge, TN | Feb 2003 – Aug 2004 https://saic.com

Maintained and expanded the functionality of a legacy Perl application used for network-wide resource and account maintenance at the Oak Ridge and Los Alamos National Labs until it was replaced. **Tech:** Perl, C, Oracle, Unix

University of Tennessee – Center for Environmental Biotechnology

Knoxville, TN | Aug 2002 – Oct 2003 https://ceb.utk.edu/ Built bioinformatics tools for monitoring, reporting, and analysis of microarray gene data. Tutored professional staff and graduate students in Perl programming for bioinformatics research.**Tech:** Perl, C, MySQL, Red Hat Linux

Manhattan Associates

Atlanta, GA | Jul 2001 – Jul 2002

Performed code maintenance on a cross-platform warehouse management application written in C++ for Linux and Windows, with Linux as the primary platform. **Tech:** C++, Oracle, Linux, Windows, MKS Toolkit, Microsoft Advanced Windows Server OS

LHS/Sema Group Telecom

Atlanta, GA | Jan 2000 – Jul 2001

Performed code maintenance and enhancement for a large legacy billing engine written in C/C++ using Oracle, spanning multiple Unix platforms. Contributed to major technical documentation rewrites. **Tech:** C, C++, Oracle, Unix

Intellivoice / Premier Technologies

Atlanta, GA | Nov 1998 – Jan 2000

Worked on multimedia integration R&D for distributed Solaris networks. Developed coding projects using C++ and evaluated third-party middleware (such as CORBA) to facilitate network communication. **Tech:** C++, CORBA, Solaris

MCI/WorldCom

Atlanta, GA | Feb 1996 – Nov 1998

Full-cycle project to implement a billing engine on AIX using C++, Pro*C, and Oracle. Developed a Perl application that processed firearm purchase applications for the State of California (an MCI client). **Tech:** C++, SQL, Unix, Perl